

BREMEN, Germany

WMO: 102240

Lat: 53.0450N

Lon: 8.7978E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-6.8	-12.2	1.3	-7.2	-10.0	1.6	-5.8	13.9	7.1	12.7	7.2	3.9	80	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	29.8	19.6	27.3	19.0	25.4	18.1	20.8	27.2	19.8	25.7	18.9	24.0	4.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.6	23.2	17.9	12.8	22.0	17.0	12.1	21.3	59.8	27.4	56.6	25.8	53.7	24.1	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.5	8.5	-11.6	33.1	3.2	2.0	-13.9	34.6	-15.8	35.8	-17.6	36.9	-20.0	38.3		
(5)			-12.0	22.7	3.2	0.7	-14.3	23.2	-16.2	23.6	-18.0	24.0	-20.3	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	1.7	2.4	5.1	9.5	12.8	16.0	18.6	17.9	14.5	10.4	5.9	3.2
	DBStd	6.97	4.51	4.14	3.84	3.91	3.83	3.41	3.40	3.07	2.92	3.45	3.72	4.54
	HDD10.0	1077	257	214	156	57	14	1	0	0	2	37	129	212
	HDD18.3	3192	515	447	409	266	176	87	39	46	121	245	372	469
	CDD10.0	1034	1	1	4	41	102	181	266	245	137	50	7	1
	CDD18.3	107	0	0	0	6	17	46	32	6	0	0	0	0
	CDH23.3	984	0	0	0	8	56	162	445	268	43	2	0	0
	CDH26.7	274	0	0	0	8	34	148	77	7	0	0	0	0
(14) Wind	WSAvg	4.2	5.0	4.8	4.8	4.2	4.0	3.9	3.8	3.6	3.7	3.9	4.2	4.6
(15) Precipitation	PrecAvg	712	52	43	40	57	69	62	81	81	55	59	61	64
	PrecMax	879	88	94	62	104	124	151	151	168	184	150	95	130
	PrecMin	566	12	12	18	12	21	22	32	14	14	22	15	9
	PrecStd	104	25	25	15	27	28	38	27	42	35	32	23	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	13.1	18.1	24.8	27.2	29.9	33.2	31.9	27.2	22.2	15.9	13.1
		MCWB	10.4	9.2	12.0	14.9	18.5	20.0	20.5	20.3	19.1	16.8	13.6	11.3
	2%	DB	10.8	10.9	15.0	21.6	24.8	27.2	30.3	28.9	24.1	19.0	14.0	11.9
		MCWB	9.3	8.6	10.4	13.4	16.6	19.2	19.7	19.3	18.1	15.6	12.6	10.3
	5%	DB	9.0	9.4	12.6	19.1	22.6	25.1	28.0	26.1	21.8	17.0	12.9	10.5
		MCWB	8.0	7.9	9.5	12.7	15.5	18.2	19.3	18.4	16.7	14.5	11.4	9.3
	10%	DB	7.8	8.1	10.9	16.7	20.1	22.8	25.8	24.0	19.8	15.4	11.2	9.0
		MCWB	6.9	6.7	8.4	11.7	14.4	17.0	18.6	17.7	15.7	13.5	9.6	8.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.2	12.9	15.7	19.7	21.5	22.1	21.7	20.2	17.3	14.3	11.7
		MCDB	12.2	11.2	16.9	21.3	26.4	27.6	28.6	28.4	26.0	21.1	15.4	13.0
	2%	WB	9.5	9.0	11.1	14.4	17.9	19.9	20.9	20.3	18.4	15.9	12.7	10.7
		MCDB	10.5	10.5	13.7	19.8	23.2	25.9	28.0	26.2	23.0	18.5	13.8	11.7
	5%	WB	7.9	7.9	10.0	13.3	16.4	18.7	20.0	19.3	17.2	14.8	11.4	9.4
		MCDB	8.7	9.3	12.3	18.1	20.8	23.3	26.3	24.5	20.8	16.7	12.7	10.3
	10%	WB	6.6	6.6	8.7	12.1	14.9	17.6	19.1	18.3	16.1	13.7	10.0	7.9
		MCDB	7.4	7.8	10.7	16.1	19.0	21.6	24.6	22.8	18.8	15.5	11.0	8.8
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	7.5	9.6	9.7	9.6	9.9	9.8	8.7	7.3	5.1	4.4
		MCDBR	5.2	6.7	10.8	13.3	13.0	13.2	13.6	13.3	11.9	9.2	5.9	4.9
		MCWBR	4.8	5.3	7.0	7.1	6.7	6.6	5.5	5.7	6.4	6.2	4.8	4.5
	5% WB	MCDBR	4.9	6.1	9.5	11.7	11.6	11.8	12.2	10.9	10.2	8.2	5.8	4.9
		MCWBR	4.7	5.2	6.6	6.8	6.8	6.6	5.5	5.5	6.0	5.8	5.0	4.7
(40) Clear-Sky Solar Irradiance	taub		0.327	0.346	0.378	0.413	0.411	0.409	0.425	0.417	0.390	0.362	0.341	0.319
	taud		2.327	2.322	2.248	2.201	2.246	2.273	2.243	2.280	2.344	2.395	2.356	2.317
	Ebn at Noon		643	750	797	815	839	843	820	804	780	723	618	576
	Edh at Noon		61	82	108	128	128	126	128	117	98	76	59	52
(44) All-Sky Solar Radiation	RadAvg		0.60	1.28	2.40	3.98	4.92	5.19	4.97	4.14	3.01	1.70	0.74	0.46
	RadStd		0.06	0.18	0.26	0.48	0.54	0.41	0.66	0.30	0.36	0.17	0.08	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+1.15	+1.78	N/A	N/A	N/A	N/A	-151	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A

Nomenclature: See separate page